

STEFAN STRAINOVIC | CV

➤ MSc in Applied Mathematics (Statistics) with experience in finance, oncology and clinical-style modeling. I'm used to turning analyses into usable tools and collaborating with researchers and engineers. Looking to apply rigorous stats to real world machine learning.

📁 EXPERIENCES

Credit Risk ML Engineer (Python) Karmen, July 2024 - January 2025

- **Credit Risk Model Development:** Researched, built, and integrated ML models for loan risk assessment, achieving high 90%+ accuracy that enabled real-time assessments.
- **Transaction Classification:** Designed a TF-IDF pipeline to train client-specific classification and clustering models, rapidly labeling thousands of transactions per client.
- **Document Intelligence & Tooling:** Explored CNN/TF-IDF approaches for unknown bank statements; delivered internal UIs, dashboards, and fixes to improve team productivity.

Research Assistant (Oncology) (R) ENSICAEN, March 2023 - July 2023

- **Statistical Analysis:** Conducted analyses of T-lymphocyte (white blood cell) sub-types following X-ray and proton brain irradiation.
- **Modeling:** Built multiple regression models to study relationships between treatment variables and lymphocyte concentrations, highlighting irradiation effects.
- **Findings & Interpretation:** Analyzed results and drew conclusions on therapy impact to inform optimization of treatment variables.

Project Manager/Producer (Volunteer) Gazamo Ltd., 2011 - 2016

- **Leadership:** Led and coordinated multiple video-game map design projects with an international artist team.
- **Planning & Delivery:** Developed project scopes and timelines, ensuring milestones were met within budget and schedule.
- **Hands-on Contribution:** Contributed directly to environment design and iteration to meet project objectives.

📖 EDUCATION

Master of Applied Mathematics (Statistics) University of Caen, October 2024

- Focus on developing complex statistical models, applying quantitative methods for decision making, and orchestrating data governance, cleaning, and analysis workflows.
- Extracurricular Activities: Volunteered at the ESN as (Erasmus Student Network) student association as a regional representative and a event organiser.

Bachelor of Mathematics University of Paris Diderot, July 2022

- Comprehensive training in fundamental and applied mathematics, spanning analysis, algebra, probability, and numerical methods.
- Interdisciplinary studies in art, linguistics and language acquisition.

📁 PROJECTS & HACKATHONS (H)

3D Brain Tumor Segmentation (nnU-Net, GASA)

- Multi-class MRI segmentation (background, necrotic, edema, enhancing). nnU-Net backbone with a GASA attention branch (axial 2D projections → MHSA → fusion) for long-range context, patch-based training, Dice+CE loss, BraTS-style per-class Dice/NSD.

(H) Predicting Cognitive Decline (Ensemble BERT + MLP)

- Extracted prosodic/audio features (pitch, jitter/shimmer, pause metrics) and transcripts; ran sentiment/lexical features, trained an ensemble with BERT on text and an MLP on audio & sentiment to classify decline.

(H) Bacteriophage Genomic Classifier

- Built a token level classifier mapping gene segments to phage components (head/neck/tail), curated FASTA data, normalized labels, performed sequence tokenization and fine-tuned a transformer (Mistral) on genomic tokens.

Brain Tumor Classifier (CNN, 2D slices)

- 98% accuracy on 2D MRI slices via transfer learning (VGG16/VGG19), data augmentation and cross-validation, reproducible training/evaluation scripts.

CONTACT

📍 Paris, France

☎ +33 6 95 02 23 14

✉ stefan.strain@outlook.com

🔗 <https://stefanstrain.github.io/>

🌐 github.com/StefanStrain

www.linkedin.com/in/stefanstrainovic

LANGUAGES

English - C2 (IELTS 7.5)

●●●●●●●●

Serbian - C2 (Native)

●●●●●●●●

French - C1/C2

●●●●●●●●

PROGRAMMING LANGUAGES AND TOOLS

Python

R

SAS

SQL

AWS (S3, SageMaker, EC2)

Github

Docker | Kubernetes

LaTeX

Excel | VBA

Power BI

DATA SCIENCE AND ML LIBRARIES

pandas | NumPy | SciPy | polars

matplotlib | seaborn | plotly

scikit-learn | XGBoost | LGBM

CatBoost | PyTorch | TensorFlow

Transformers | SHAP | statsmodels

NiBabel | SageMaker

MLflow | Optuna | Hyperopt

Streamlit | FastAPI | Docker

spaCy | Whisper | tokenizers

APPLIED STATISTICS SKILLS

Exploratory data analysis (EDA)

Regression & generalized linear models

Mixed-effects & hierarchical models

Time series & multivariate methods

Supervised learning

Unsupervised learning

Feature engineering & leakage control

Model evaluation & calibration

Model monitoring: drift & performance

Causal inference & Survival analysis

Experimental design & A/B testing

Trial design, randomization & power

Missing data & sensitivity analyses